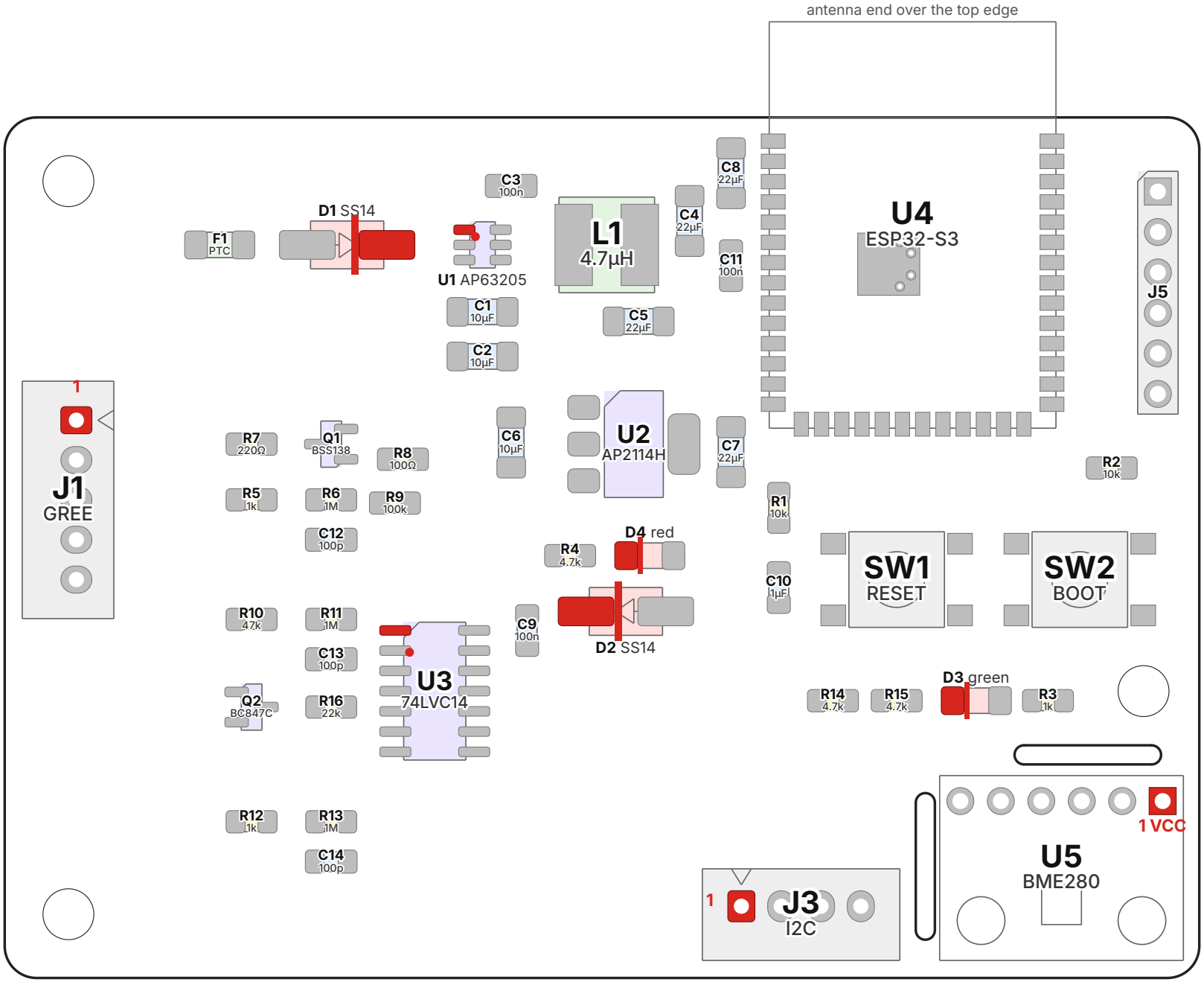
# Gree bridge rev B: parts placement and orientation

Top side (all parts on top) · scale 2.8 : 1 · gree_bridge.kicad_pcb · **red = check the direction before soldering**




**Diodes and LEDs:** red band + red pad = CATHODE. SS14: band on the body. LEDs: the stripe under the clear lens marks the cathode end (or meter diode test).

**ICs:** red dot = pin 1 (dot or bevel on the chip). **Connectors, BME280:** red "1" = pin 1 (square pad). XH headers: deep side to the board edge.

Body colours: resistors · capacitors · diodes / LEDs · ICs / transistors · inductor / fuse

# Gree bridge rev B: assembly checklist

One row per Mouser part number, in soldering order: low and fine-pitch parts first, through-hole last.

| Done | Step | Refs | Part | Mouser | Qty | Direction / how to place |
|---|---|---|---|---|---|---|
| ☐ | 1 | **U1** | AP63205 buck 5V TSOT-23-6 | 621-AP63205WU-7 | 1 | **Pin-1 dot on the red dot. No dot: text readable, pin 1 is bottom-left.** |
| ☐ | | **U3** | 74LVC14 Schmitt SOIC-14 | 595-SN74LVC14ADRG4 | 1 | **Pin-1 dot / bevelled edge on the red dot (top-left).** |
| ☐ | 2 | **U2** | AP2114H LDO 3.3V SOT-223 | 621-AP2114H-3.3TRG1 | 1 | Fits one way: wide tab on the wide pad. |
| ☐ | 3 | **U4** | ESP32-S3-WROOM-1 N16R8 | 356-ESP32S3WRM1N16R8 | 1 | Antenna end over the top edge. Centre pad from the back (below). |
| ☐ | 4 | **R1, R2** | 10k ohm 1% 0805 | 603-RC0805FR-0710KL | 2 | Either way. |
| ☐ | | **R3, R5, R12** | 1k ohm 1% 0805 | 603-RC0805FR-071KL | 3 | Either way. |
| ☐ | | **R4, R14, R15** | 4.7k ohm 1% 0805 | 603-RC0805FR-074K7L | 3 | Either way. |
| ☐ | | **R6, R11, R13** | 1M ohm 1% 0805 | 603-RC0805FR-071ML | 3 | Either way. |
| ☐ | | **R7** | 220 ohm 1% 0805 | 603-RC0805FR-07220RL | 1 | Either way. |
| ☐ | | **R8** | 100 ohm 1% 0805 | 603-RC0805FR-07100RL | 1 | Either way. |
| ☐ | | **R9** | 100k ohm 1% 0805 | 603-RC0805FR-07100KL | 1 | Either way. |
| ☐ | | **R10** | 47k ohm 1% 0805 | 603-RC0805FR-0747KL | 1 | Either way. |
| ☐ | | **R16** | 22k ohm 1% 0805 | 603-RC0805FR-0722KL | 1 | Either way. |
| ☐ | 5 | **C1, C2, C6** | 10uF 25V X7R 1206 | 603-CC126KKX7R8BB106 | 3 | Either way. |
| ☐ | | **C3, C9, C11** | 100nF 50V X7R 0805 | 603-CC805KRX7R9BB104 | 3 | Either way. |
| ☐ | | **C4, C5, C7, C8** | 22uF 25V X5R 1206 | 810-C3216X5R1E226M | 4 | Either way. |
| ☐ | | **C10** | 1uF 25V X7R 0805 | 603-CC805KKX7R8BB105 | 1 | Either way. |
| ☐ | | **C12, C13, C14** | 100pF 50V C0G 0805 | 603-CC805JRNPO9BN101 | 3 | Either way. |
| ☐ | 6 | **F1** | PTC fuse 0.5A 24V 1206 | 530-0ZCJ0050AF2E | 1 | Either way. |
| ☐ | | **L1** | 4.7uH inductor 6×6mm | 810-VLS6045EX-4R7M | 1 | Either way. |
| ☐ | 7 | **D1, D2** | SS14 Schottky 40V 1A SMA | 512-SS14 | 2 | **Band = cathode, on the red band. D1 band toward U1, D2 band to the left.** |
| ☐ | | **D3** | LED green 1206 | 859-LTST-C150KGKT | 1 | **Cathode on the red band: lens-stripe end, or check with the diode test.** |
| ☐ | | **D4** | LED red 1206 | 859-LTST-C150KRKT | 1 | **Cathode on the red band: lens-stripe end, or check with the diode test.** |
| ☐ | 8 | **Q1** | BSS138 N-MOSFET SOT-23 | 512-BSS138 | 1 | Fits one way. Q1 (BSS138) and Q2 (BC847C) look alike: keep apart. |
| ☐ | | **Q2** | BC847C NPN SOT-23 | 863-BC847CLT1G | 1 | Fits one way. Q1 (BSS138) and Q2 (BC847C) look alike: keep apart. |
| ☐ | 9 | **SW1, SW2** | Tact switch 6×6mm SMD | 611-PTS645SL43SMTR92 | 2 | Legs on the 4 pads: fits only the right way (either 180° turn). |
| ☐ | 10 | **J1** | JST XH 5-pin vertical THT | 306-B5BXHALFSN | 1 | **Square pad = pin 1. Deep side of the body to the board edge.** |
| ☐ | | **J3** | JST XH 4-pin vertical THT | 306-B4BXHALFSN | 1 | **Square pad = pin 1. Deep side of the body to the board edge.** |
| ☐ | | **J5** | Pin header 1×6 2.54mm THT | 710-61300611121 | 1 | Symmetric. Short pin ends into the board. |
| ☐ | 11 | **U5** | GY-BME280 breakout (yours) | - | 1 | **Short pins in the breakout, long pins through U5. Chip up, VCC square.** |

- U4 centre pad: it has a 1.0 mm plated hole open on the back. Solder the side pads, then heat the hole from the back and feed solder until it flows through (or use hot air).
- First power-up from J5 only (5 V through D2): check 5 V and 3.3 V before connecting J1.
- C8 has no label on the board (next to the module's top-left corner).